STAT 7900: Special Topic: Causal Analysis



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STAT 7900

The four assumptions — big picture

What they mean • why we need them • what happens when they fail

Consistency

Exchangeability

Positivity

SUTVA / No interference

These assumptions are what allow an observed-data analysis to be interpreted causally.

Four Causal Assumptions

1. Consistency

The observed outcome equals the potential outcome under the treatment received.

2. Exchangeability

After conditioning on the right covariates, treatment assignment carries no extra information about the potential outcomes.

3. Positivity

Within relevant covariate strata, every treatment option has a nonzero chance of being observed.

4. SUTVA / no interference

Treatment is well defined, and one person's treatment does not change another person's outcome.

All four matter. Different assumptions fail in different ways.

Assumption 1: Consistency



If the person receives treatment 1, then the observed outcome should equal that person's potential outcome under treatment 1.

Consistency connects the observed outcome Y to the counterfactual notation $Y(t)$.

Assumption 1: Consistency

Question: Does taking 10 mg of Drug A reduce systolic blood pressure after 4 weeks?

Treatment definition

T = 1 means:
- Drug A
- 10 mg daily
- for 4 weeks

Why consistency is plausible

People labeled T = 1 are receiving essentially the same intervention that defines Y(1).

The clearer the intervention, the easier consistency is to defend.

Assumption 1: Consistency

Walking
20 min/week

Running
5 days/week

Strength
training

Mixed
program

All four people are coded T = 1, but these are not the same intervention.

If treatment versions matter for Y, then “Y(1)” is ambiguous.

Assumption 2: Exchangeability

Can the treated and untreated groups stand in for each other's counterfactual outcomes?

Ideal comparison

Apart from treatment, the groups are comparable with respect to factors that determine the outcome.

Why we need it

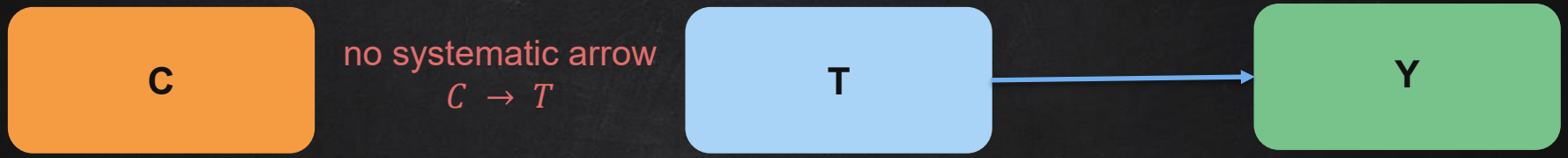
We use untreated people to learn what would have happened to treated people without treatment, and vice versa.

No unblocked confounding

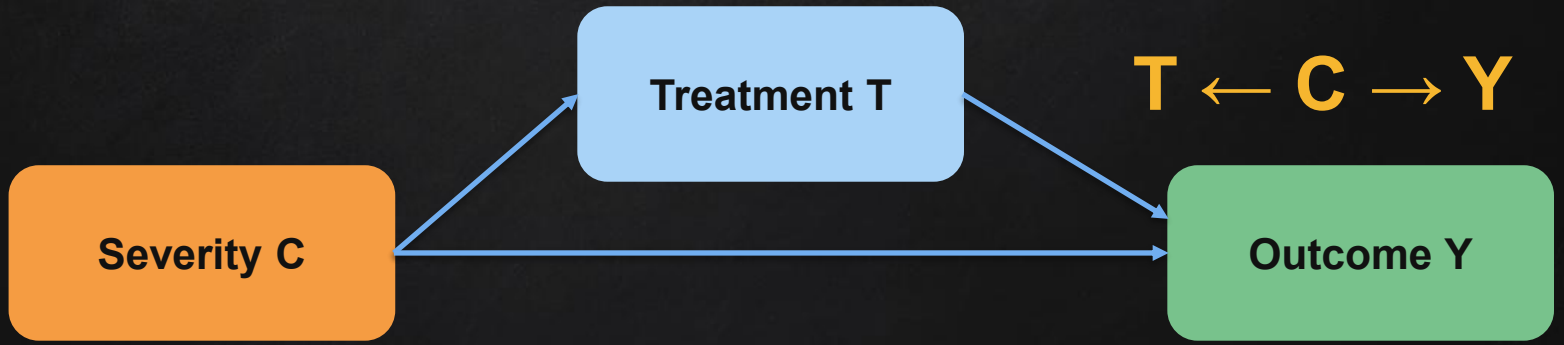
Exchangeability is the assumption that makes between-person comparisons informative about counterfactuals.

Assumption 2: Exchangeability

Exchangeability in a randomized trial



Exchangeability in observational data



Conditional Exchangeability

Observational studies usually rely on exchangeability after conditioning on X.

$$Y(t) \perp T \mid X$$

Interpretation

Within people with the same relevant X, treatment behaves “as if randomized” with respect to the potential outcomes.

Critical limitation

This only works if X actually contains enough information to block the important backdoor paths.

This is exactly why DAGs matter: they help justify what belongs in X.

Assumption 3: Positivity

For every kind of person we want to compare, we need some treated and some untreated observations.

$$0 < P(T = t \mid X = x) < 1$$

Plain language

Within each relevant covariate stratum, treatment cannot be impossible or guaranteed.

Why we need it

Without both treatment options represented, the missing counterfactual must be extrapolated rather than learned from data.

Positivity is an overlap requirement.

Assumption 3: Positivity

Example: patients with a contraindication NEVER receive the drug.

Stratum X = contraindication

$P(T = 1 \mid X) = 0$

There are no treated patients in this stratum.

Consequence

The treated potential outcome for this group is unsupported by observed data.

More data will not fix a structural zero.

Positivity can fail because of design, clinical rules, geography, policy, or extreme covariate combinations.

Assumption 3: Positivity

Structural vs practical positivity problems

Structural violation

Treatment truly cannot occur in a stratum.

Example: pregnancy is an absolute contraindication for a specific drug.

Practical violation

Treatment is possible but extremely rare.

Example: 2 treated patients versus 500 untreated patients in a covariate region.

The assumption is theoretical; overlap diagnostics are empirical clues.

Assumption 4: SUTVA / No Interference

SUTVA is easiest to understand as two separate ideas.

A. No interference

One person's treatment does not change another person's outcome.

B. No hidden treatment versions

"Treatment 1" corresponds to a sufficiently well-defined intervention.

SUTVA tells us whose treatment can affect whose outcome — and what exactly "treatment" means.

Assumption 4: SUTVA / No Interference

Example: vaccination in an infectious-disease study



A's vaccination may reduce B's infection risk.

Then B's outcome depends not only on B's own treatment, but also on A's treatment.

Assumptions summary

Consistency

Does observed Y correspond to Y(t)?

Exchangeability

Are treatment groups comparable?

Positivity

Do both treatment options exist across X?

SUTVA

Is treatment well defined, without interference?

If any assumption fails, the causal interpretation can fail.

STAT 7900 - Quiz 2



Why DAG?

The potential-outcomes assumption leaves us with a practical question.

$$Y(t) \perp T \mid X$$

Within levels of X, treatment assignment carries no additional information about the potential outcomes.

BUT WHAT BELONGS IN X?

Age? Baseline severity? Mediator? Every measured variable?

A DAG helps us decide which variables to adjust for, and which variables NOT to adjust for.

Three Building Blocks

CHAIN

$$X \rightarrow Z \rightarrow Y$$

Open by default.
Blocked when conditioned on Z.

FORK

$$X \leftarrow Z \rightarrow Y$$

Open by default.
Blocked when conditioned on Z.

COLLIDER

$$X \rightarrow Z \leftarrow Y$$

Blocked by default.
Open when conditioned on Z.

Conditioning behaves differently in the collider.

d-Separation ≠ Causal Adjustment

d-SEPARATION

Question:

Can we make T and Y conditionally independent?

Goal: BLOCK ALL paths.

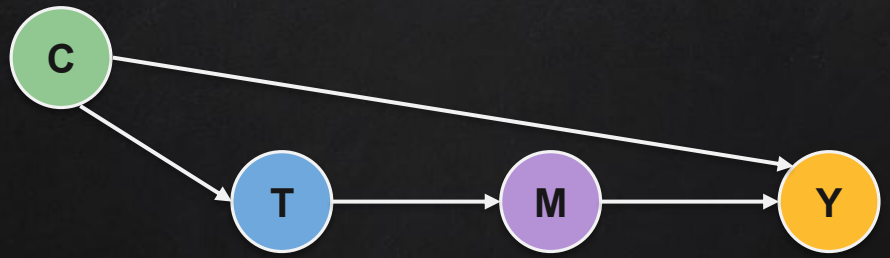
CAUSAL ADJUSTMENT: TOTAL EFFECT

Question:

What should we condition on to identify the total effect?

Goal: BLOCK NONCAUSAL backdoor paths.

Example:



Backdoor path:
T ← C → Y

Causal path:
T → M → Y

For total effect: adjust for C, not M.

d-Separation vs Causal Adjustment

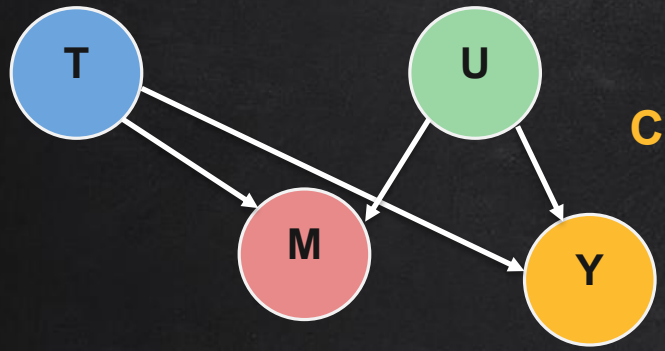
d-separation

- Goal: conditional independence
- Block all open paths between two nodes
- Useful for reading conditional independences from a DAG

Causal adjustment

- Goal: identify the total causal effect
- Block noncausal backdoor paths
- Preserve causal paths from T to Y

Collider adjustment bias with randomized T



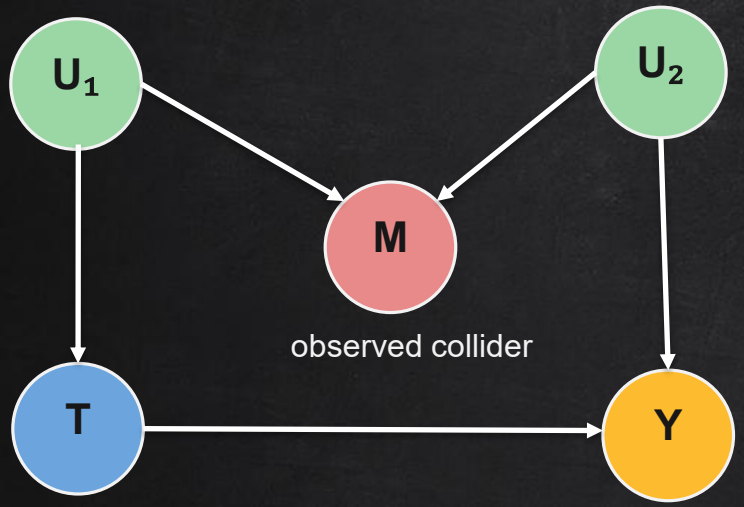
Conditioning on M opens: $\mathrm{T} \rightarrow \mathrm{M} \leftarrow \mathrm{U} \rightarrow \mathrm{Y}$

Example:

- T: randomized chemotherapy
- M: treatment-induced weight loss
- U: underlying frailty
- Y: survival

If you adjust for weight loss M, you induce an association between treatment and frailty, and that can bias the treatment effect on survival.
Randomization protects you at baseline, but you can destroy that protection by conditioning on the wrong post-treatment variable.

M-bias



Without adjusting for M, the path
$T \leftarrow U_1 \rightarrow M \leftarrow U_2 \rightarrow Y$ is blocked at M.

- Not every predictor of treatment or outcome is a confounder.
- A variable can be statistically associated with both T and Y yet still be a bad adjustment variable.
- Real-world covariate choice should combine subject-matter knowledge, temporal ordering, and causal structure.

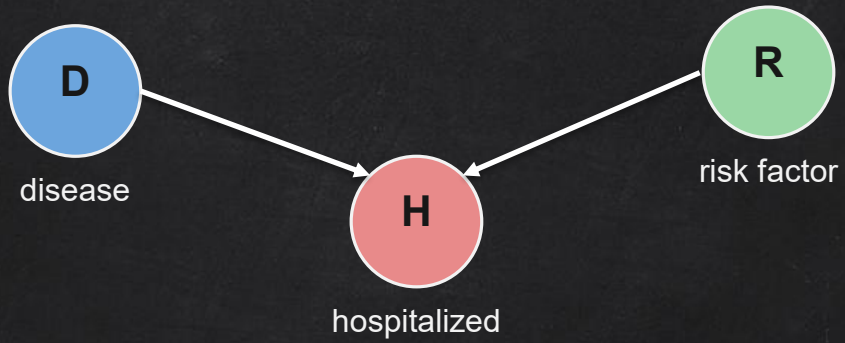
Overadjustment Bias

Overadjustment usually means conditioning on a variable that lies on, or is caused by, a causal pathway you intend to estimate.



Adjust for M → estimated "effect of T" no longer contains T → M → Y.

Selection Bias



If your dataset contains only H = 1 patients, you have conditioned on a collider.

Inside the hospital, D and R can appear associated even if they are independent in the population.

How to draw a DAG from a research question

1. Define treatment/exposure T and outcome Y clearly.

2. List causes of T, causes of Y, and common causes.

3. Add plausible mediators and important selection variables.

4. Respect temporal ordering whenever possible.

5. Draw arrows from cause to consequence, not from correlation to correlation.

6. Use the graph to inspect backdoor paths and identify adjustment sets.

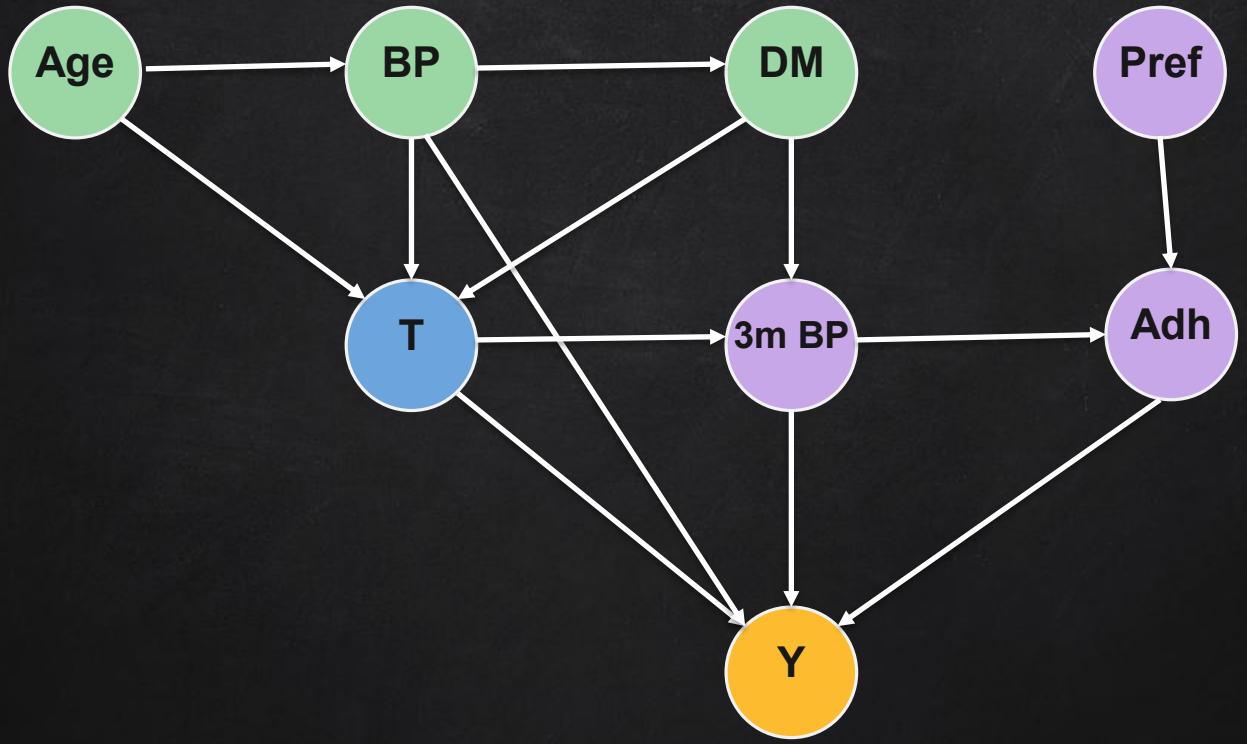
Build a graph

Research question: Does a new intensive blood-pressure treatment reduce 1-year cardiovascular events?

Candidate variables:
Age • baseline systolic BP • diabetes • physician treatment preference • treatment assignment • achieved BP at 3 months • medication adherence • cardiovascular event

Draw plausible DAG. Circle the variables you would adjust for to estimate the TOTAL treatment effect.

One plausible DAG



First: Specify the Estimand

What exactly are we trying to estimate from this DAG?

Assume our target is the TOTAL effect of treatment T on cardiovascular outcome Y.

Total Effect Means ...

- Keep every causal pathway that begins at treatment T and ends at Y.
- Block open backdoor paths that begin with an arrow into T.
- Do not automatically adjust for post-treatment variables.
- Do not automatically adjust for variables simply because they predict Y.

The estimand comes first; the adjustment set follows from the estimand + DAG.

Classify the variables by time

Baseline / pre-treatment

- Age
- Baseline BP
- Diabetes (DM)
- Physician preference

Post-treatment

- 3-month BP
- Adherence
- Cardiovascular outcome Y

Causal paths from T to Y

Path 1 $T \rightarrow Y$

Path 2 $T \rightarrow 3m\ BP \rightarrow Y$

Path 3 $T \rightarrow 3m\ BP \rightarrow Adh \rightarrow Y$

These are the pathways we want to KEEP OPEN when the target is the total effect.

Open Backdoor paths from T to Y

Path 4 T ← BP → Y

Path 5 T ← BP → DM → 3m BP → Y

Path 6 T ← BP → DM → 3m BP → Adh → Y

Path 7 T ← DM ← BP → Y

Path 8 T ← DM → 3m BP → Y

Path 9 T ← DM → 3m BP → Adh → Y

Path 10 T ← Age → BP → Y

Path 11 T ← Age → BP → DM → 3m BP → Y

Path 12 T ← Age → BP → DM → 3m BP → Adh → Y

Every backdoor path starts with an arrow INTO T. We need to block all of them, not just one.

A collider blocked T to Y path

Path 13 $T \rightarrow 3m\ BP \leftarrow DM \leftarrow BP \rightarrow Y$

Why is it closed?

3m BP is a collider: $T \rightarrow \mathbf{3m\ BP} \leftarrow \mathbf{DM}$

Without conditioning on 3m BP	Condition on 3m BP (or descendant Adh)
Path 13 is CLOSED	**Path 13 can OPEN**

So 3m BP is dangerous twice: it is a mediator on causal paths and a collider on Path 13.

Finding minimal adjustment set

Goal Block every open noncausal path from T to Y — while leaving the causal pathways open.

Step 1

Find a path that forces a variable into the adjustment set.

Step 2

See which other backdoor paths are automatically blocked.

Step 3

Add another variable only if an open backdoor path remains.

Minimal = every included variable is necessary; remove one and at least one backdoor path reopens.

Finding minimal adjustment set

Path 4

$T \leftarrow BP \rightarrow Y$

Open backdoor path. BP is the only internal node.

Adjust for BP

To block this path, there is only one option:

Why can’t we use Age or DM to block Path 4?

Because neither Age nor DM lies on this particular path. Conditioning on an off-path variable does not block the path.

Therefore, every valid standard adjustment set for this DAG must contain BP.

Once we adjust for BP, many paths disappear

Path 4	T ← BP → Y	BLOCKED by BP
Path 5	T ← BP → DM → 3m BP → Y	BLOCKED by BP
Path 6	T ← BP → DM → 3m BP → Adh → Y	BLOCKED by BP
Path 7	T ← DM ← BP → Y	BLOCKED by BP
Path 10	T ← Age → BP → Y	BLOCKED by BP
Path 11	T ← Age → BP → DM → 3m BP → Y	BLOCKED by BP
Path 12	T ← Age → BP → DM → 3m BP → Adh → Y	BLOCKED by BP

After adjusting for BP, what remains open?

Only two backdoor paths do not contain BP:

Path 8

$$T \leftarrow DM \rightarrow 3m\ BP \rightarrow Y$$

BP does not appear on this path, so BP cannot block it.

Path 9

$$T \leftarrow DM \rightarrow 3m\ BP \rightarrow Adh \rightarrow Y$$

Again, BP is not on the path.

So, BP alone is NOT sufficient.

We need one more baseline variable to close Paths 8 and 9.

DM closes the remaining two paths

Only two backdoor paths do not contain BP:

Path 8 $T \leftarrow DM \rightarrow 3m\ BP \rightarrow Y$

Path 9 $T \leftarrow DM \rightarrow 3m\ BP \rightarrow Adh \rightarrow Y$

Condition on DM

Both paths are blocked immediately at DM.

{ BP, DM }

Why not block using 3-month BP or adherence?

Mathematically, conditioning on 3m BP or Adh could block those individual paths.

But they are post-treatment variables.

3-month BP

$$T \rightarrow 3m\ BP \rightarrow Y$$

Mediator on a causal pathway

Adherence

$$T \rightarrow 3m\ BP \rightarrow Adh \rightarrow Y$$

Also lies downstream of treatment

For the TOTAL effect, block backdoor paths using baseline variables, not mediators.

Why isn’t Age required?

Age creates backdoor paths, but every Age path passes through BP:

Path 10 $T \leftarrow \text{Age} \rightarrow \text{BP} \rightarrow Y$

Path 11 $T \leftarrow \text{Age} \rightarrow \text{BP} \rightarrow \text{DM} \rightarrow \text{3m BP} \rightarrow Y$

Path 12 $T \leftarrow \text{Age} \rightarrow \text{BP} \rightarrow \text{DM} \rightarrow \text{3m BP} \rightarrow \text{Adh} \rightarrow Y$

Because BP is already in the adjustment set, all three paths are already blocked.

Age may be included, but it is not necessary for a minimal sufficient set.

Valid vs minimal adjustment sets

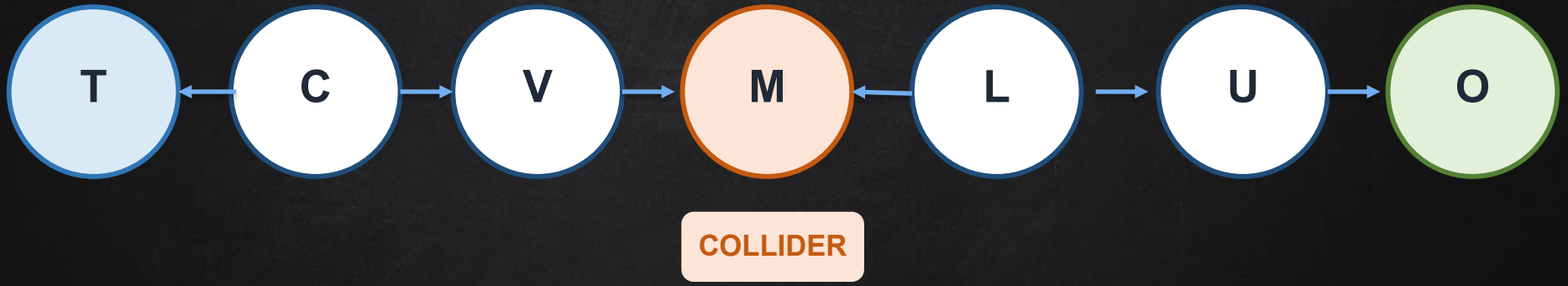
Candidate set	Valid?	Reason
{BP}	No	Paths 8 and 9 remain open
{DM}	No	Path 4 remains open
{Age, BP}	No	Paths 8 and 9 remain open
{BP, DM}	**YES — minimal**	**Blocks all open backdoor paths**
{Age, BP, DM}	**Yes, but not minimal**	Age is redundant once BP is included

Minimal does not mean “fewest variables possible” in general; it means no variable in the chosen valid set is redundant.

Collider logic is path-specific

Example 1: A backdoor path with a collider

Path: T ← C → V → M ← L → U → O

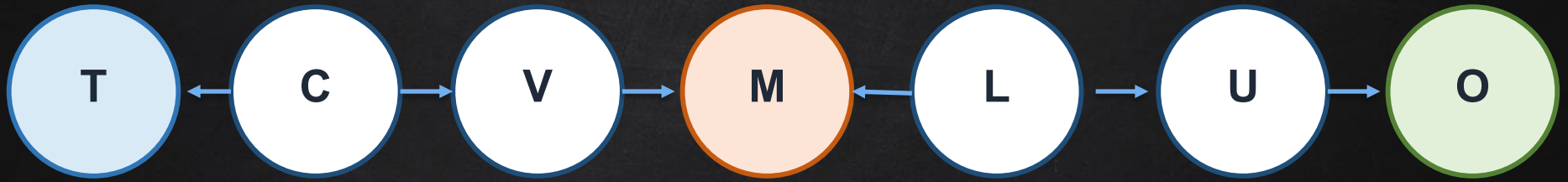


No adjustment is needed to close this particular path.

Collider logic is path-specific

Example 1: What if we condition on C?

Path: T ← C → V → M ← L → U → O



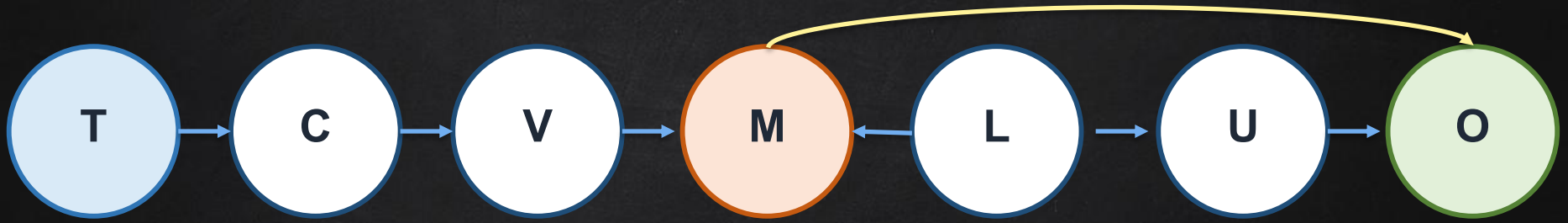
C is a non-collider on this path.
Conditioning on C blocks the path at C.

M is still an unconditioned collider.
So the path is also closed at M.

Result: the path remains CLOSED.

Collider logic is path-specific

Example 2: Now suppose we have arrow from M -> 0



Path A: T ← C → V → M → O

Path B: T ← C → V → M ← L → O

Same graph. Same node M. Different path roles.

Collider logic is path-specific

Path A: T ← C → V → M → O

Path is open since M is not a collider.

So to close it we need to condition on C.

Path B: T ← C → V → M ← L → U → O

M is a collider on this path, so the path is closed. No need to condition.

Collider present and unconditioned → CLOSED by default.

Key Lesson

- **Confounder / mediator / collider are path roles, not permanent variable identities.**
- **One node may play different roles on different paths.**
- **Adjustment decisions are made relative to a specific estimand and a specific DAG.**
- **Temporal ordering is necessary information, but it is not sufficient.**

What did the graph buy us?

Last class: $Y(t) \perp T \mid X$

Today: Use a DAG to justify a set X that blocks all backdoor paths.

Then the graphical assumptions support conditional exchangeability for the causal effect of interest.

Graphical identification is the bridge from assumptions to an estimable formula.

After block backdoor paths: adjustment formula

$$E[Y(t)] = \sum_{x} E[Y \mid T = t, X = x] P(X = x)$$

$$\mathrm{ATE} = \mathrm{E}[\mathrm{Y}(1)] - \mathrm{E}[\mathrm{Y}(0)]$$

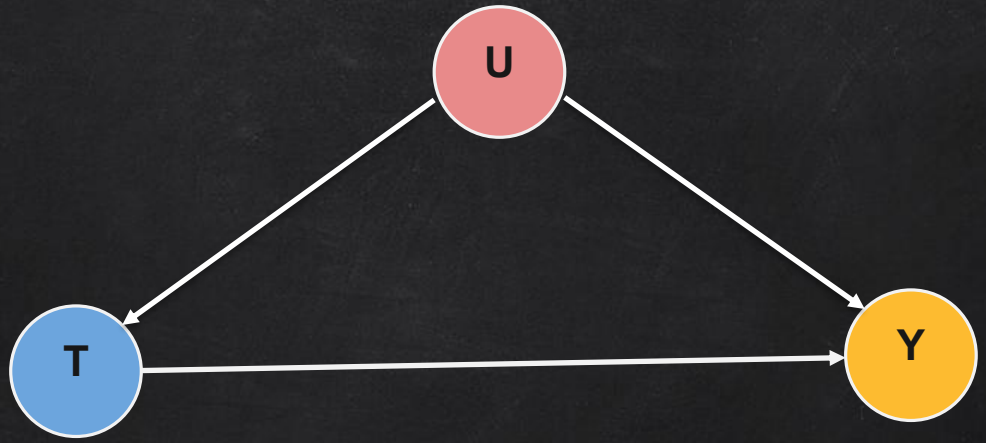
The DAG helps justify WHICH X makes this identification step valid. Regression, matching, weighting, and g-computation are estimation strategies after identification.

A DAG does not prove causality

- The arrows come from scientific knowledge, design, temporal order, and assumptions.
- Observed data generally cannot determine a unique causal DAG by itself.
- Multiple DAGs can imply the same set of observed conditional independences.
- A correct-looking picture is not a substitute for substantive justification.

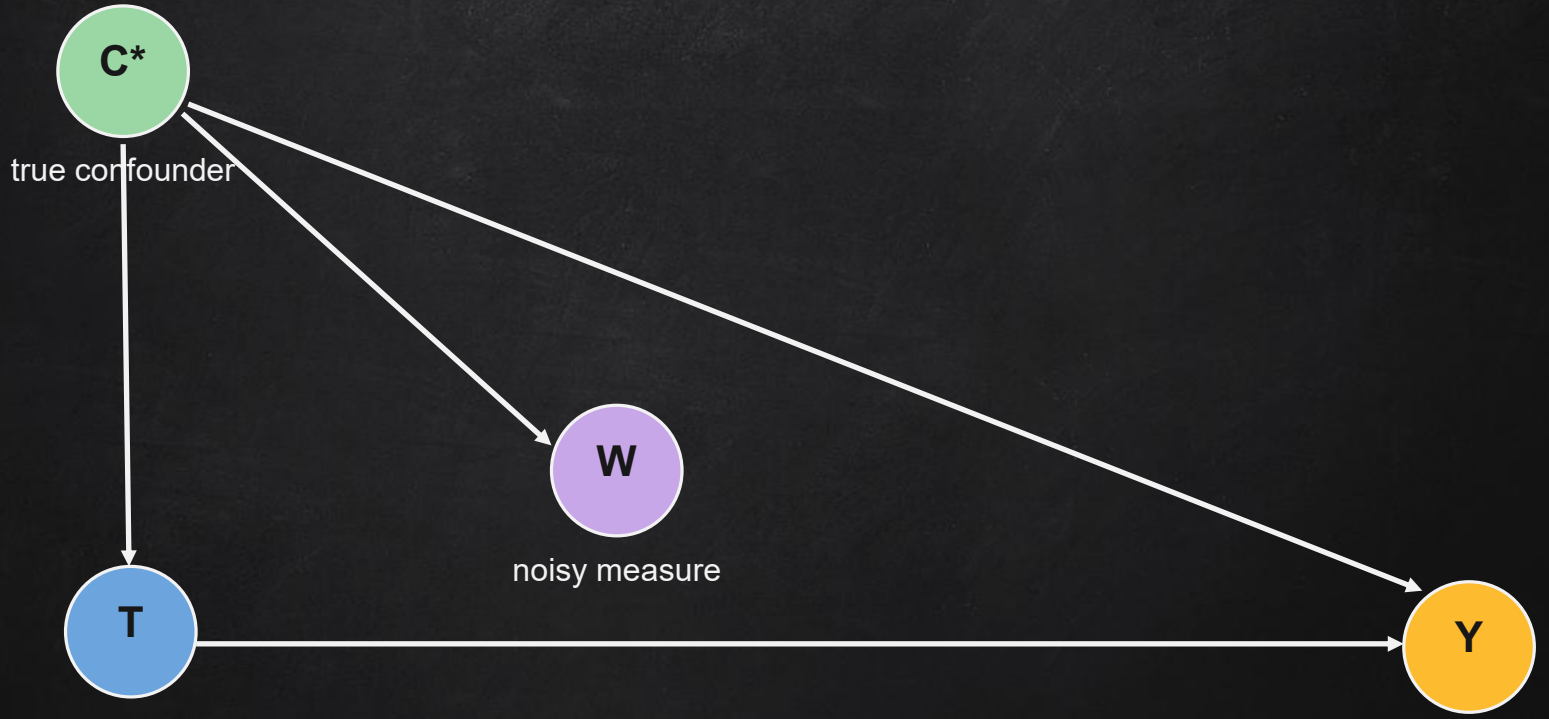
DAGs make assumptions visible; they do not make them automatically true.

Unmeasured confounding: Hidden U



If U is unmeasured, ordinary adjustment cannot block $T \leftarrow U \rightarrow Y$.

Measuring poorly is not same as measuring it



Conditioning on W may leave residual confounding because W does not perfectly recover C*.

Summary

Variable adjusted for	Graph role	What adjustment does
Confounder C	$T \leftarrow C \rightarrow Y$	Removes bias
Mediator M	$T \rightarrow M \rightarrow Y$	Blocks part of total effect
Collider C	$T \rightarrow C \leftarrow Y$	Creates association / bias

Regression syntax cannot tell you which causal role a covariate has.

Two branches of analysis

CAUSAL ANALYSIS WITH A DAG

Start with a scientifically motivated graph.

Use it to reason about identification, adjustment, and bias.

CAUSAL DISCOVERY

Use patterns of conditional independence and assumptions to learn aspects of graph structure.

Outputs may be equivalence classes (CPDAGs/PAGs), not one certain DAG.

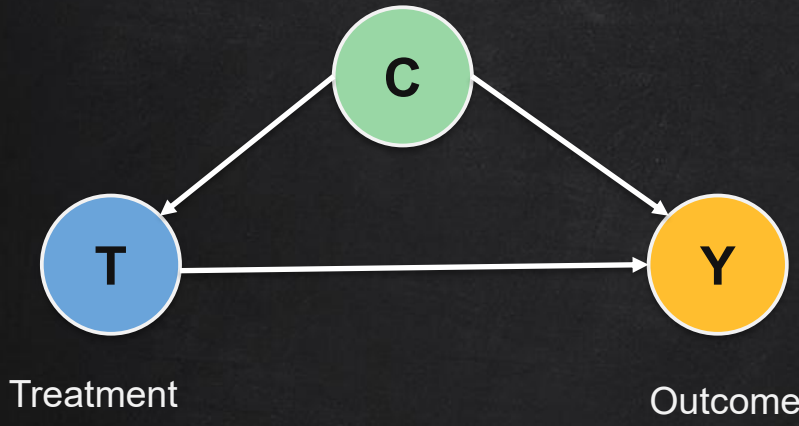
R Lab Simulation

- Confounding can make a treatment effect look too large or too small.
- Adjusting for the right backdoor variable can recover the causal effect.
- Adjusting for a mediator changes the estimand: total effect → direct effect.
- Adjusting for a collider can create bias, even when treatment is randomized.
- The same DAG can lead to different conditioning choices for d-separation versus causal adjustment.

Predict first → run the code → explain the result from the DAG.

Simulation 1

C = baseline risk




```r
set.seed(7901)
n <- 5000
C <- rnorm(n)
T <- rbinom(n, 1, plogis(C))
Y <- 2*T + 1.5*C + rnorm(n)
```

Backdoor path: $T \leftarrow C \rightarrow Y$

True treatment effect in the simulation = 2

Will the crude model overestimate or underestimate the effect?

Simulation 1

```
set.seed(7901)
   n <- 5000

   C <- rnorm(n)
T <- rbinom(n, 1, plogis(C))
Y <- 2*T + 1.5*C + rnorm(n)

   fit_crude <- lm(Y ~ T)
 fit_adj   <- lm(Y ~ T + C)

coef(fit_crude)["T"] - 3.18
 coef(fit_adj)["T"] - 1.99
```

MODEL 1

$Y \sim T$

Backdoor path remains open.

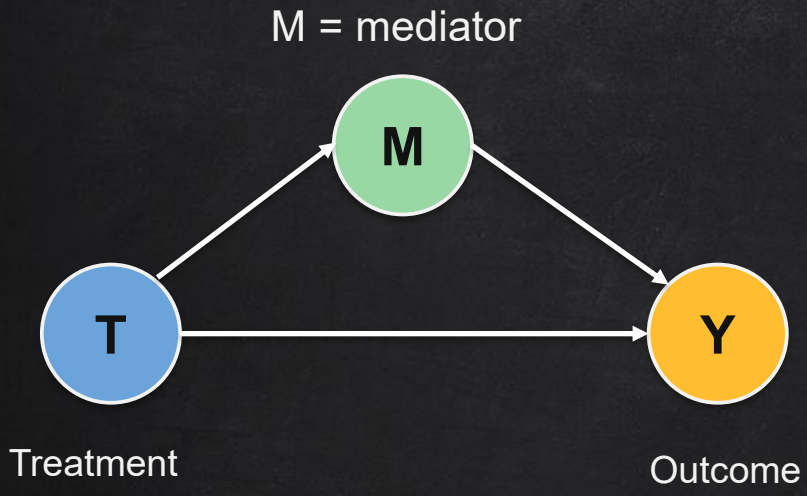
MODEL 2

$Y \sim T + C$

Backdoor path is blocked.

Adjusted coefficient ≈ 2

Simulation 2




```r
set.seed(7902)
n <- 5000
T <- rbinom(n, 1, 0.5)
M <- 1.2*T + rnorm(n)
Y <- 1.0*T + 2.0*M + rnorm(n)
```

$T \rightarrow M \rightarrow Y$ is part of the causal effect.

Target total effect? Do NOT condition on M.

Simulation 2

```
set.seed(7902)
   n <- 5000

   T <- rbinom(n, 1, 0.5)
   M <- 1.2*T + rnorm(n)
Y <- 1.0*T + 2.0*M + rnorm(n)

   fit_total  <- lm(Y ~ T)
 fit_direct <- lm(Y ~ T + M)

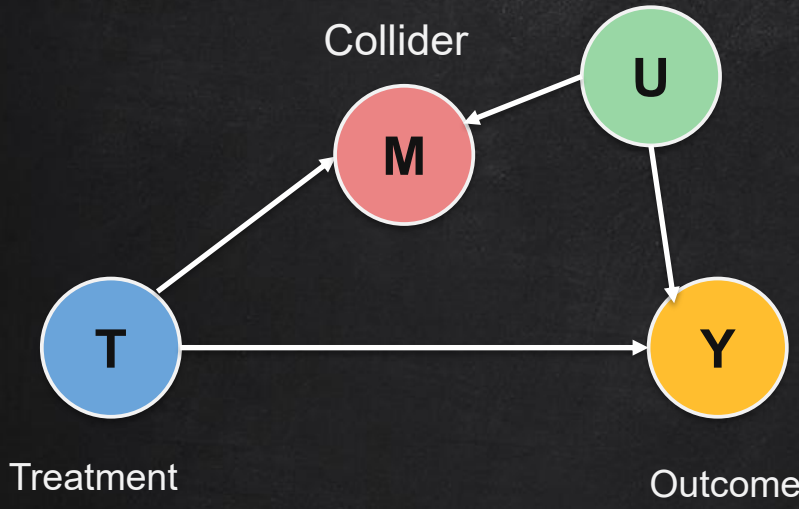
coef(fit_total)["T"]   - 3.38
coef(fit_direct)["T"] -  0.98
```

Without M
T coefficient ≈ 3.4
TOTAL EFFECT

With M
T coefficient ≈ 1.0
DIRECT EFFECT

Adding a covariate can change the scientific question.

Simulation 3




```r
set.seed(7903)
n <- 5000
T <- rbinom(n, 1, 0.5)
U <- rnorm(n)
M <- 1.2*T + 1.2*U + rnorm(n)
Y <- 1.5*U + rnorm(n)
```

Before conditioning: $T \rightarrow M \leftarrow U \rightarrow Y$ is CLOSED at M.

Condition on M → the collider path OPENS.

Simulation 3

```
set.seed(7903)
  n <- 5000

  T <- rbinom(n, 1, 0.5)
  U <- rnorm(n)
M <- 1.2*T + 1.2*U + rnorm(n)
Y <- 1.5*U + rnorm(n)   # no effect of T

  fit_unadj <- lm(Y ~ T)
fit_col   <- lm(Y ~ T + M)

coef(fit_unadj)["T"] - 0.009
coef(fit_col)["T"] - -0.867
```

$$Y \sim T$$

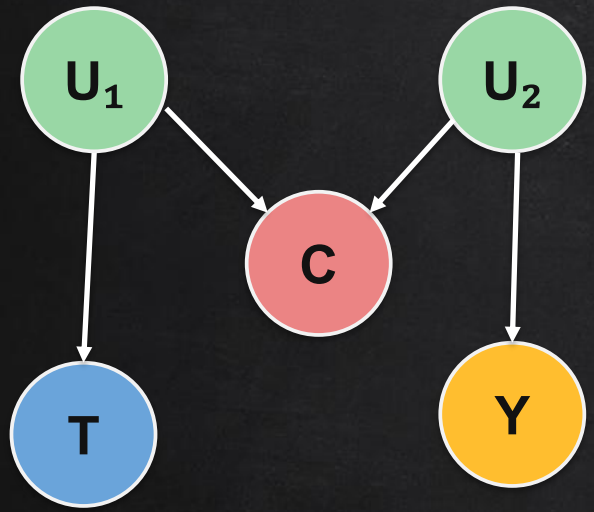
T coefficient $\approx$ 0

$$Y \sim T + M$$

T coefficient moves away from 0

Randomization does not protect against bad post-treatment adjustment.

Simulation 4




```r
set.seed(7904)
n <- 100000
U1 <- rnorm(n)
U2 <- rnorm(n)
T <- rbinom(n, 1, plogis(0.9*U1))
C <- 0.9*U1 + 0.9*U2 + rnorm(n)
Y <- 1.5*U2 + rnorm(n)
```

Path $T \leftarrow U_1 \rightarrow C \leftarrow U_2 \rightarrow Y$ is CLOSED at C.

Adjusting for C OPENS the path.

Simulation 4

```
set.seed(7904)
  n <- 100000

U1 <- rnorm(n)
U2 <- rnorm(n)
T  <- rbinom(n, 1, plogis(0.9*U1))
C  <- 0.9*U1 + 0.9*U2 + rnorm(n)
Y  <- 1.5*U2 + rnorm(n)

fit0 <- lm(Y ~ T)
fit1 <- lm(Y ~ T + C)

coef(fit0)["T"] : 0.01
coef(fit1)["T"] : -0.35
```

No adjustment
T coefficient ≈ 0

Adjust for collider C
spurious $T-Y$ association appears

Timing is not enough. We need the causal role of the variable.

d-Separation vs Backdoor Adjustment

d-SEPARATION

Question:
Can we make T and Y conditionally independent?

Goal:
Block ALL paths.

BACKDOOR ADJUSTMENT

Question:
What should we condition on to identify the total effect of T on Y?

Goal:
Block NONCAUSAL backdoor paths; preserve causal paths.

Summary

Topic	Structure	Rule
Confounder	$T \leftarrow C \rightarrow Y$	Usually adjust
Mediator	$T \rightarrow M \rightarrow Y$	Do not adjust for total effect
Collider	$T \rightarrow M \leftarrow U \rightarrow Y$	Do not condition
M-bias	$T \leftarrow U_1 \rightarrow C \leftarrow U_2 \rightarrow Y$	Pre-treatment does not imply adjust
d-separation	Block all paths between X and Y	Goal: conditional independence
Backdoor adjustment	Block noncausal paths into T	Goal: identify total causal effect

From Adjustment to Estimation



We now know WHAT to adjust for. How do we use the observed data to estimate the causal effect?

$$E[Y(t)] = \Sigma_x \, E[Y \mid T = t, X = x] \, P(X = x)$$

The DAG justifies WHICH X makes this identification step credible.

Next question: Is the regression coefficient of T automatically the ATE? Only in special cases.

g-computation not same as regression

Simple additive model:

$$E(Y \mid T, X) = \beta_0 + \beta_1 T + \beta_2 X$$

If the model is additive in T, then the contrast for any individual is:

$$E(Y \mid T = 1, X) - E(Y \mid T = 0, X) = \beta_1.$$

That means every person has the same model-based treatment effect. So, averaging across people gives the same β_1 again.

Therefore: g-computation ATE = regression coefficient of T.

g-computation not same as regression

Clinical example

Outcome model:

$$E(Y \mid T, X) = 140 + 0.5X - 5T - 0.2(TX)$$

T = 1: antihypertensive drug; T = 0: no drug

Y = post-treatment systolic blood pressure

X = Age − 50

Suppose the effect of treatment depends on age.

Because of the interaction term $T \times X$, the treatment effect is no longer constant across patients.

g-computation not same as regression

At $X = 0$ (that is, Age = 50):

$$E(Y \mid T = 1, X = 0) - E(Y \mid T = 0, X = 0) = -5$$

So the coefficient of T (−5) is the treatment effect only at the reference value X = 0.

It is NOT automatically the average treatment effect in the study population.

For a person with covariate value X, the model-based treatment effect is:

$$\Delta(X) = [140 + 0.5X - 5 - 0.2X] - [140 + 0.5X] = -5 - 0.2X.$$

What does g-computation do?

Step 1: fit an outcome model $E(Y \mid T, X)$.

Step 2: for each person i, predict $\hat{Y}_i(1)$ by setting $T = 1$.

Step 3: for the same person, predict $\hat{Y}_i(0)$ by setting $T = 0$.

Step 4: compute the individual contrast $\hat{Y}_i(1) - \hat{Y}_i(0)$.

Step 5: average these contrasts over the whole sample.

$$\boldsymbol{ATE_g = (1/n)\,\Sigma\,[\,\hat{Y}_i(1) - \hat{Y}_i(0)\,]}$$

Simple Illustration I

Patient	Age	X = Age−50	Predicted effect = −5 − 0.2X
A	50	0	−5
B	60	10	−7
C	70	20	−9

g-computation average = (−5 − 7 − 9)/3 = −7

Simple Illustration II

Suppose the study population has mean age 65.

Then mean $X = 65 - 50 = 15$.

Average model-based treatment effect:

$$ATE = E[-5 - 0.2X] = -5 - 0.2E[X] = -5 - 0.2(15) = -8.$$

Regression coefficient of T = −5

g-computation ATE = −8

These differ because treatment effect varies with X.

g-computation vs regression

Regression coefficient of T is a conditional effect tied to the chosen parameterization.

g-computation targets a marginal causal estimand: average over the covariate distribution.

They are equal in special cases, especially a simple additive linear model with no interaction.

They generally differ with interactions, nonlinear models, logistic regression, splines, or machine learning.

Therefore: g-computation is NOT “just the T coefficient”.

regression = model
g-computation = model + counterfactual prediction + averaging

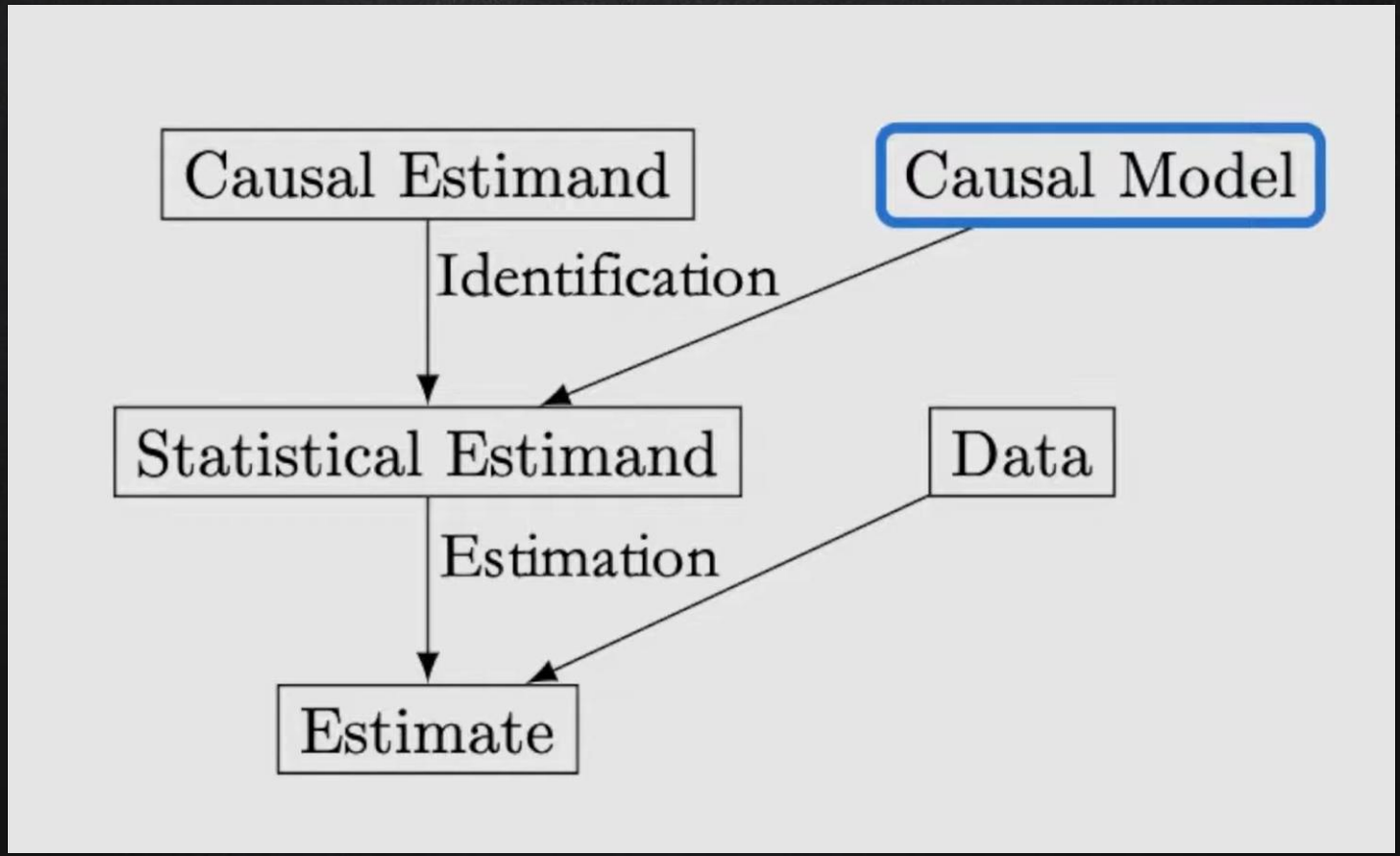
Short R workflow

```
fit <- lm(Y ~ T * X, data = dat)
  dat1 <- transform(dat, T = 1)
  dat0 <- transform(dat, T = 0)
y1hat <- predict(fit, newdata = dat1)
y0hat <- predict(fit, newdata = dat0)
   ate_g <- mean(y1hat - y0hat)
```

Notice: we didn’t use coef(fit)["T"] as the ATE.

We predicted both counterfactual outcomes for everyone and then averaged.

Structural Causal Models (SCM)



Structural Causal Models (SCM)

From a causal picture to a model of how the world generates variables

DAG = causal map

SCM = causal map + generating rules

We will learn how to represent an intervention such as “force treatment = 1.”

Structural Causal Models (SCM)

A DAG already tells us:

- which variables directly cause which.
- which paths are causal or noncausal.
- what we may need to adjust for.

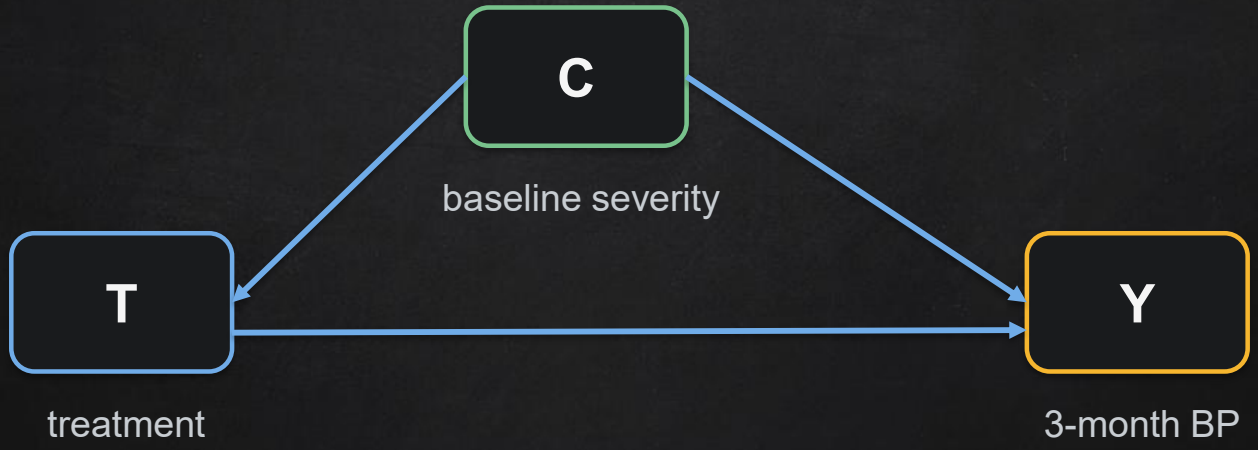
SCM adds something new:

- a rule for how each variable is generated.
- a precise meaning for “intervene”.
- a way to define the same person under different treatment choices.

SCM makes “what if we changed T?” mathematically precise.

Example

Clinical question: **Does an intensive blood-pressure treatment lower systolic BP after 3 months?**



Example

$C \rightarrow T$ Baseline severity can influence which treatment is chosen.

$C \rightarrow Y$ Baseline severity can also influence the outcome.

$T \rightarrow Y$ Treatment can causally change the outcome.

The DAG gives direction, but not yet the actual rule for generating T or Y.

Example

The graph does not tell us:

- How strongly does severity affect treatment choice?
- How strongly does treatment change blood pressure?
- What other background factors affect treatment or outcome?
- Exactly what should we change when we say “give treatment to everyone”?

SCM supplies those generating rules.

SCM

1
Generate baseline severity C

2
Generate treatment T using C + other factors

3
Generate outcome Y using T, C + other factors

Each recipe says: “given the direct causes of this variable, how is this variable produced?”

U Variables

U means background factors not explicitly represented as nodes in our simple DAG.

U_C	U_T	U_Y
genetics, long-term history, measurement noise	clinician preference, logistics, patient choice	other causes of the outcome

We use U because no model includes every cause in the universe.

Important: a U variable can be measured or unmeasured in reality. Here it simply means “outside the simplified system we drew.”

Treatment Assignment

$$T := f_T(C, U_T)$$

This does NOT mean treatment is random.

It says treatment is produced by some rule that can depend on baseline severity C and other treatment-related factors U_T.

f_T = Treatment Decision Rule

Outcome

$$Y = 120 + 10C - 6T + U_Y$$

One possible numerical rule:

- Every 1-unit increase in baseline severity raises Y by 10 units.
- Treatment lowers Y by 6 units, holding the other causes fixed.
- U_Y represents other causes of Y not shown explicitly.

What makes an SCM causal

A regression can describe association.

An SCM makes an additional scientific claim:

If we externally change one variable, the other generating rules remain the rules of the system.

That is what allows the model to answer intervention questions.

Key Operation: Intervention

Normal observational world:

$$T = f_T(C, U_T)$$

Treatment follows the usual decision process.

Intervention world:

$$T := 1$$

We replace the treatment rule and force treatment to 1.

Everything else in the SCM stays the same.

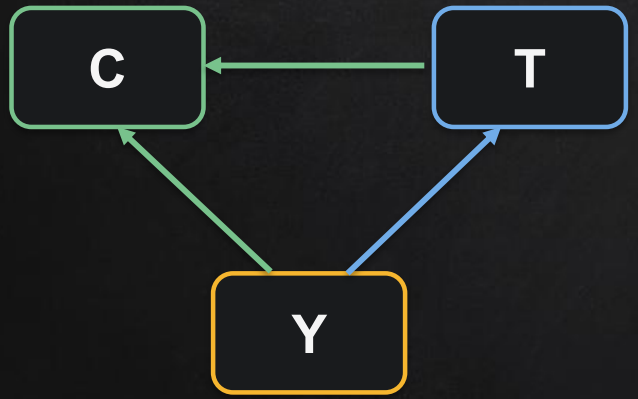
Do Calculus

“Set treatment to 1 from outside the system.”

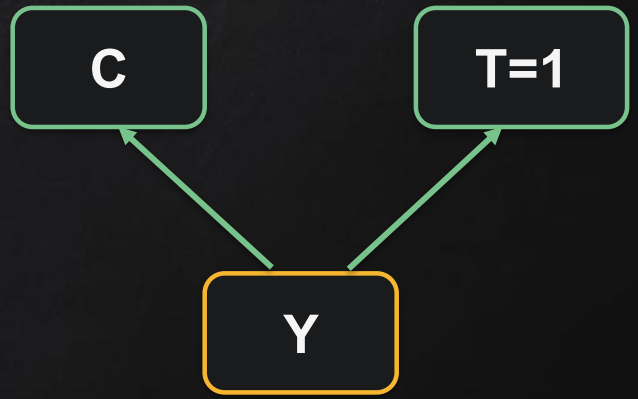
$$do(T = 1)$$

- Do not let baseline severity decide T.
- Do not let clinician preference decide T.
- Simply assign T = 1, then let the rest of the model run.

Before intervention



After do(T = 1)



Intervention vs Conditioning

Conditioning: T = 1
Look at the people who happened to receive treatment.

Their baseline characteristics may differ from untreated people.

Intervention: do(T = 1)
Force treatment to 1 for the target population.

The usual treatment-selection process is bypassed.

$P(Y \mid T = 1)$ and $P(Y \mid do(T = 1))$ answer different questions.

Numerical Example

Outcome rule: $Y = 120 + 10C - 6T$

Patient A

$C = +1\ (sicker)$

$T = 1$

$Y = 124$

Patient B

$C = -1\ (healthier)$

$T = 0$

$Y = 110$

Observed comparison: $124 - 110 = +14$

But treatment itself lowers Y by 6. Baseline severity created the misleading observed difference.

Three Different Questions

1. Association
What is Y among people we observe with T=1 versus T=0?

$$P(Y \mid T = t)$$

2. Intervention
What would Y be if we forced T=t for the target population?

$$P(Y \mid do(T = t))$$

3. Counterfactual
For this same person, what would Y have been under another treatment?

$$Y(t)$$

SCM gives one framework for all three, but they are not the same question.

Counterfactual: Same person, Different treatment

Suppose one person has: $C = 1.5 \quad U_Y = 2$

Keep these person-specific background factors fixed.

World 1: $T = 1$

$$Y(1) = 120 + 10(1.5) - 6 + 2 = 131$$

World 0: $T = 0$

$$Y(0) = 120 + 10(1.5) + 2 = 137$$

$$\mathrm{Y}(1) - \mathrm{Y}(0) = -6$$

g-Computation does the same thing

1
set T = 1 and predict Y

2
set T = 0 and predict Y

3
average the difference

That procedure is trying to emulate the two intervention worlds described by the SCM.

The estimation model is not the SCM itself, but the intervention logic is the same.

Simulation: Build Data from SCM

$$C \sim \text{Normal}(0,1)$$

$$P(T=1 \mid C) = \text{logistic}(1.2C)$$

$$Y = 120 + 10C - 6T + \varepsilon$$

Interpretation:

- Sicker patients (larger C) are more likely to receive treatment.
- Sicker patients also have higher Y.
- Treatment truly lowers Y by 6 units.

True intervention effect = −6

Simulation: Build Data from SCM

```
set.seed(7900)
n <- 5000

C <- rnorm(n)
p <- plogis(1.2*C)
T <- rbinom(n, 1, p)

eps <- rnorm(n, sd = 5)
Y <- 120 + 10*C - 6*T + eps

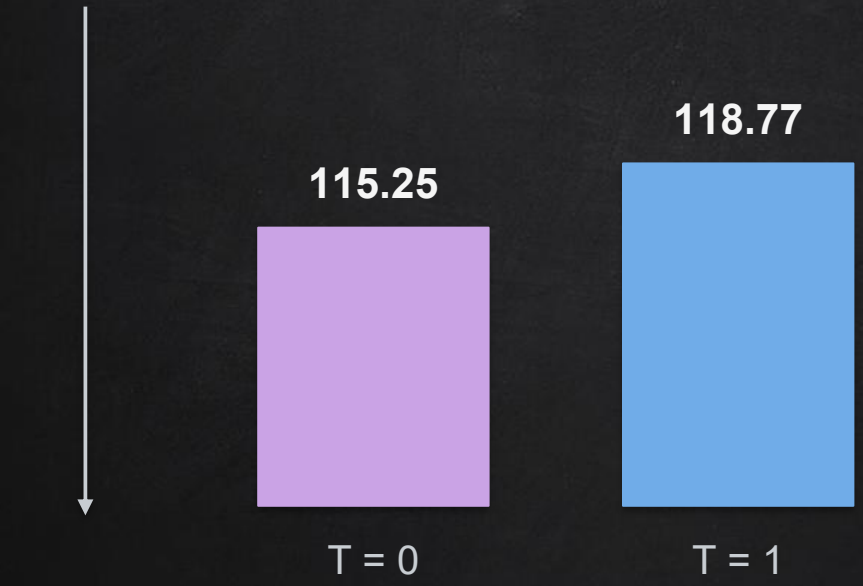
dat <- data.frame(C, T, Y)
```

Before fitting anything, predict:

- Will treated and untreated patients have the same average C?
- Will the crude difference in Y recover -6?
- What should happen after we adjust for C?

Simulation: Build Data from SCM

Observed mean Y



Crude observed difference:

$$118.77 - 115.25 = +3.52$$

But the true treatment effect is −6.

Wrong sign!

One simulated realization; exact sample numbers vary slightly with the random-number generator.

Simulation adjust for Common Cause C

```
  fit_crude <- lm(Y ~ T, data = dat)
fit_adj   <- lm(Y ~ T + C, data = dat)

        coef(fit_crude)["T"]
            # about +3.5

         coef(fit_adj)["T"]
            # about -6.0
```

Crude model
T coefficient ≈ +3.5

Adjusted model
T coefficient ≈ −6.0

The adjusted model recovers the truth here because the simulation satisfies the required assumptions and the outcome rule is correctly represented.

Simulation creates two interventions

```
# keep each person's C and eps fixed
    Y1 <- 120 + 10*C - 6*1 + eps
    Y0 <- 120 + 10*C - 6*0 + eps

     mean(Y1)      # about 114.0
     mean(Y0)      # about 120.0
      mean(Y1 - Y0) # exactly -6
```

do(T = 1)
mean Y ≈ 114.0

do(T = 0)
mean Y ≈ 120.0

$$\text{ATE} = 114 - 120 = -6$$

Real data example: job training and later earnings

Causal question: Does participation in the National Supported Work job-training program increase earnings in 1978?

LaLonde / MatchIt data

614	185	429
people	treated	controls

Treatment T: job-training participation
Outcome Y: 1978 earnings (re78)

What was known before treatment?

Age • education • race • marital status
No high-school degree • earnings in 1974 • earnings in 1975

Call these baseline variables X

These variables may affect both program participation and later earnings.

What does SCM add?

Treated	Control	Crude difference
$6,349	$6,984	−$635

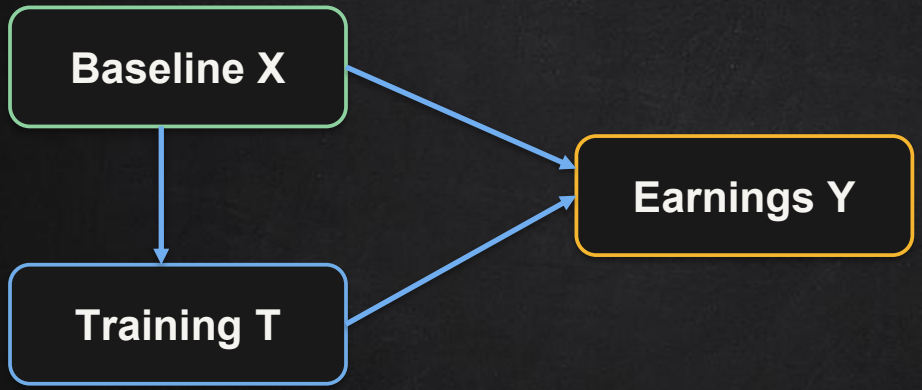
But the groups were already different before treatment

Baseline variable	Treated	Control
Age	25.8	28.0
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619
1975 earnings	$1,532	$2,466

So −$635 is an observed association, not automatically the causal effect of training.

What do observed data say?

A plausible causal story



What we observed

$$E[Y \mid T = 1] - E[Y \mid T = 0] = -\$635$$

What we want causally

$$E[Y \mid do(T = 1)] - E[Y \mid do(T = 0)]$$

SCM tells us what “force training” versus “force no training” means. It does not make −$635 causal; assumptions and appropriate adjustment are still needed.

Next Lesson

SCM gave us a precise target:

$$E[Y \mid do(T=1)] - E[Y \mid do(T=0)]$$

Next question:

When can we learn that intervention effect from the data we actually observe?

Next: randomized experiments and identification